

## Lecture 10

# The periodic Toda chain: spectral curve and transfer matrix

**Objectives.** Close the open chain of [Lecture 9](#) into a ring. Its Lax matrix becomes a tight-binding Hamiltonian, and Bloch's theorem puts a variable  $z$  into its corners. The same eigenvalue problem, as a first-order recursion, gives a  $2 \times 2$  Lax matrix at every site, a discrete zero-curvature equation, and the monodromy matrix of [Lecture 8](#), whose trace  $t(\lambda)$  generates commuting integrals. The ring is Liouville integrable, with tori once its drift is split off, and all its integrals sit in one curve,  $z + 1/z = t(\lambda)$ : bands, gaps, and two spectral parameters that read the same curve two ways.

## 10.1 Periodic boundary conditions

Put  $N$  particles on a ring and let particle  $N$  interact with particle 1 across the join ([Figure 10.1](#)). With the positions read cyclically,  $q_{n+N} = q_n$ ,<sup>1</sup> the Hamiltonian is

$$H = \frac{1}{2} \sum_{n=1}^N p_n^2 + \sum_{n=1}^N e^{q_n - q_{n+1}}, \quad q_{N+1} = q_1, \quad (10.1)$$

the open chain (9.3) with one more spring,  $e^{q_N - q_1}$ . The equations of motion are those of the open chain with every index read modulo  $N$ :  $\dot{q}_n = p_n$ ,  $\dot{p}_n = e^{q_{n-1} - q_n} - e^{q_n - q_{n+1}}$ . For  $N = 3$ , in centre-of-mass coordinates and after a rescaling, this is the Hamiltonian (6.8) whose Poincaré section [Figure 6.9](#) showed smooth curves where Hénon–Heiles had chaos. This lecture proves that what the figure showed is integrability.

### The Lax pair of the ring

Flaschka's variables (7.10) carry over with one more bond,  $a_N = e^{(q_N - q_1)/2}$ , and one new constraint: the exponents telescope to zero round the ring, so

$$a_1 a_2 \cdots a_N = 1. \quad (10.2)$$

Put  $a_N$  in the two corners:

$$L = \begin{pmatrix} b_1 & a_1 & & & a_N \\ a_1 & b_2 & a_2 & & \\ & \ddots & \ddots & \ddots & \\ & & a_{N-2} & b_{N-1} & a_{N-1} \\ a_N & & & a_{N-1} & b_N \end{pmatrix}, \quad (10.3)$$

<sup>1</sup>Physically, particle  $n+N$  is particle  $n$  displaced once round the ring: its position is  $Q_{n+N} = Q_n + \ell$ . The variables  $q_n := Q_n - n\ell/N$  are periodic, and every spring  $e^{Q_n - Q_{n+1}}$  becomes  $g e^{q_n - q_{n+1}}$  with the same constant  $g = e^{-\ell/N}$ ; rescaling  $p = \sqrt{g} \tilde{p}$  and  $t = \tilde{t}/\sqrt{g}$  then returns the equations below with  $g = 1$ . Nothing is lost.

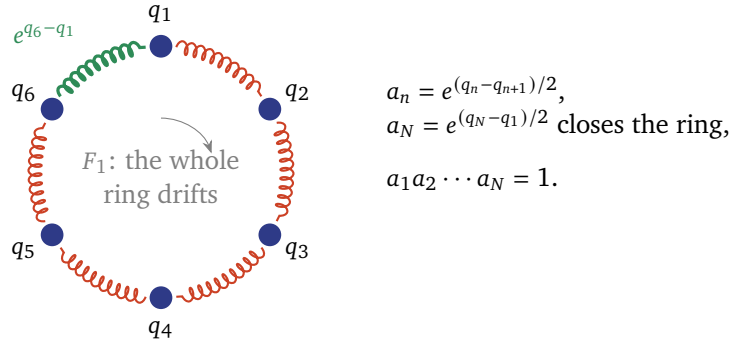


Figure 10.1: The periodic Toda chain,  $N = 6$ . The new spring (green) joins particle  $N$  to particle 1. A common translation changes no spring, so the flow of  $F_1 = \sum_n p_n$  carries the whole ring along without end.

and let  $M$  be antisymmetric with  $M_{n,n+1} = -\frac{1}{2}a_n$  for every  $n$ , indices modulo  $N$ , so that  $M_{N1} = -\frac{1}{2}a_N = -M_{1N}$ . Each entry of  $[M, L]$  involves at most two neighbouring bonds, so the band computations of [Proposition 7.7](#) go through with indices read modulo  $N$  (for  $N = 3$  the second-neighbour terms, which cancel identically, land on first-neighbour slots and change nothing):  $\dot{L} = [M, L]$  for  $N \geq 3$ , and by [Theorem 7.2](#) the  $F_k = \text{tr } L^k$  are conserved, with  $F_1 = \sum_n p_n$  and  $F_2 = 2H$ .

Read  $L$  as a physicist would: the tight-binding Hamiltonian of an electron on a ring of  $N$  sites, on-site energies  $b_n$ , hoppings  $a_n$ . As the particles move, the electron's lattice changes shape and its energy levels do not.

*Two warnings.* The ring still drifts: the flow of  $F_1$ ,  $q_n \mapsto q_n + s$ , changes no  $a_n$ , so no level set is compact ([§ 9.5](#)); [§ 10.4](#) splits the drift off. And  $N = 2$  is *degenerate*: the corner slot  $(1, 2)$  is the off-diagonal slot, so  $L_{12} = a_1 + a_2$  and  $\text{tr } L^2 = 2H + 4$ . The formulas below are for  $N \geq 3$ ; try  $N = 3$  first.

## 10.2 Bloch–Floquet puts $z$ into $L$

The eigenvalue problem  $L\psi = \lambda\psi$  is a recursion,

$$a_{n-1}\psi_{n-1} + b_n\psi_n + a_n\psi_{n+1} = \lambda\psi_n, \quad (10.4)$$

closed up by  $\psi_0 = \psi_N$  and  $\psi_{N+1} = \psi_1$ . Now unroll the ring ([Figure 10.2](#)): extend  $a_n, b_n$  periodically and let (10.4) hold for every  $n \in \mathbb{Z}$ , an electron on a crystal whose unit cell is our ring. The ring's eigenvectors are the solutions of period  $N$ , but the crystal has more states, and Bloch's theorem (from solid-state physics) says which: the bounded solutions can be chosen with

$$\psi_{n+N} = z\psi_n \quad \text{for all } n, \quad z = e^{i\theta}, \quad (10.5)$$

$\theta$  the Bloch phase per unit cell. We impose (10.5) for every non-zero complex  $z$ , the *Bloch–Floquet multiplier*.

A solution of (10.5) is fixed by  $\psi_1, \dots, \psi_N$ . Rows 2 to  $N-1$  of (10.4) are rows of  $L$  and are unchanged. Row 1 contains  $a_0\psi_0 = a_N\psi_0 = a_N z^{-1}\psi_N$ , and row  $N$  contains  $a_N\psi_{N+1} = a_N z\psi_1$ .

So the Bloch condition turns (10.4) into  $L(z)\psi = \lambda\psi$  with

$$L(z) = \begin{pmatrix} b_1 & a_1 & & & a_N/z \\ a_1 & b_2 & a_2 & & \\ & \ddots & \ddots & \ddots & \\ & & a_{N-2} & b_{N-1} & a_{N-1} \\ a_N z & & & a_{N-1} & b_N \end{pmatrix}, \quad L(1) = L. \quad (10.6)$$

Nothing was postulated:  $L(z)$  is the crystal's Hamiltonian restricted to Bloch waves.

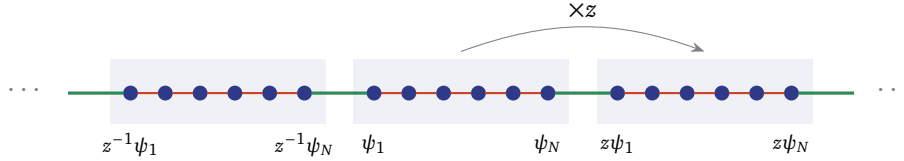


Figure 10.2: The ring unrolled into a crystal whose unit cell (shaded) is the ring. A Bloch wave repeats from cell to cell up to the factor  $z$ . Only the bonds that cross a cell boundary (green) see  $z$ , hence the corners of  $L(z)$ . The ring itself is  $z = 1$ .

**Proposition 10.1 (A Lax pair for every  $z$ ).** *Let  $M(z)$  agree with  $M$  except in the corners,  $M(z)_{N1} = -\frac{1}{2}a_N z$  and  $M(z)_{1N} = \frac{1}{2z}a_N$ . Then along every solution of (10.1),  $\dot{L}(z) = [M(z), L(z)]$  for every  $z \neq 0$ . In particular the spectrum of  $L(z)$ , and  $\det(\lambda - L(z))$  as a function of two variables, are conserved.*

The proof restricts the infinite crystal's Lax equation to Bloch waves.

Reading

*Proof of Proposition 10.1.* Let  $L_\infty$  and  $M_\infty$  be the infinite periodic tridiagonal matrices acting on sequences  $(\psi_n)_{n \in \mathbb{Z}}$ , with the entries of  $L$  and  $M$  in every cell. Entry by entry,  $\dot{L}_\infty = [M_\infty, L_\infty]$  is the open-chain computation of Proposition 7.7 with no ends at all, so it holds. Both matrices commute with the shift  $\psi_n \mapsto \psi_{n+N}$ , since their entries are periodic, so both map the space of solutions of (10.5) into itself. On that  $N$ -dimensional space, in the coordinates  $\psi_1, \dots, \psi_N$ , the derivation of (10.6) says  $L_\infty$  acts as  $L(z)$ , and the same derivation applied to  $M_\infty$  gives  $M(z)$ . Restricting the identity to the invariant subspace gives the result. ■

At  $z = e^{i\theta}$ ,  $L(z)$  is Hermitian and its eigenvalues are the Bloch energies at phase  $\theta$ ; as  $\theta$  runs round the circle they sweep out the *bands* of the crystal (§ 10.5), which by Proposition 10.1 do not change as the particles move.

*A word on names.* In § 7.9 the matrix carried  $\lambda$  and its eigenvalue was  $\eta$ . Here the matrix carries  $z$  and its eigenvalue is  $\lambda$ . That is deliberate: the eigenvalue of  $L(z)$  is the spectral parameter of a second Lax matrix, built next (§ 10.6).

### 10.3 Discrete zero curvature and plaquette holonomy

The recursion (10.4) is of second order, so it is a first-order recursion for pairs  $(\psi_{n+1}, \psi_n)$ . Solving for  $\psi_{n+1}$  means dividing by  $a_n$ , which gives a  $2 \times 2$  matrix of determinant  $a_{n-1}/a_n$  involving two bonds. A gauge transformation does better.

### The gauge that makes it local

Conjugate by  $D = \text{diag}(e^{-q_1/2}, \dots, e^{-q_N/2})$ , the gauge of Proposition 7.9 (it is § 7.8's  $D$  up to a constant). In the new variable  $\phi_n = e^{-q_n/2}\psi_n$ , (10.4) reads

$$\phi_{n+1} = (\lambda - p_n) \phi_n - e^{q_{n-1}-q_n} \phi_{n-1} : \quad (10.7)$$

the coefficient of  $\phi_{n+1}$  is now 1. (Substitute  $\psi_n = e^{q_n/2}\phi_n$  in (10.4), use  $a_n e^{q_{n+1}/2} = e^{q_n/2}$ , and divide by  $e^{q_n/2}$ .) Now collect a pair,  $\Phi_n = (\phi_n, -e^{q_{n-1}}\phi_{n-1})^T$ . Then (10.7) and the definition of the second component say exactly

$$\Phi_{n+1} = L_n(\lambda) \Phi_n, \quad L_n(\lambda) = \begin{pmatrix} \lambda - p_n & e^{-q_n} \\ -e^{q_n} & 0 \end{pmatrix}, \quad \det L_n(\lambda) = 1. \quad (10.8)$$

This is the *site matrix*: local (site  $n$  only), polynomial in the spectral parameter  $\lambda$ , and a transport one step along the lattice (Figure 10.3).

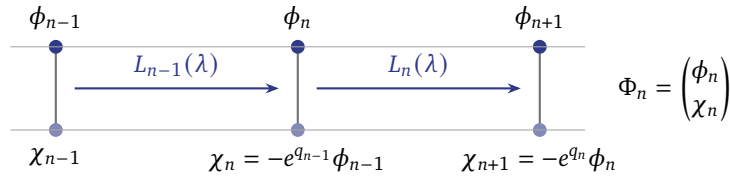


Figure 10.3: The recursion (10.7) as a ladder. Each rung is a pair  $\Phi_n$ , and the site matrix  $L_n(\lambda)$ , which contains  $(q_n, p_n)$  and nothing else, carries one rung to the next.

### Time evolution, and zero curvature

The site matrix transports  $\Phi$  in space. Transport in time is a second matrix at each site,

$$V_n(\lambda) = \begin{pmatrix} -\lambda/2 & -e^{-q_n} \\ e^{q_{n-1}} & \lambda/2 \end{pmatrix}, \quad (10.9)$$

which follows from  $\dot{\psi} = M\psi$ , the recursion and the same gauge (we omit that derivation; what matters is the next statement, checked directly).

**Proposition 10.2** (Discrete zero curvature). *For every  $\lambda$ ,*

$$\dot{L}_n(\lambda) = V_{n+1}(\lambda) L_n(\lambda) - L_n(\lambda) V_n(\lambda) \quad \text{for all } n \quad (10.10)$$

*holds if and only if  $\dot{q}_n = p_n$  and  $\dot{p}_n = e^{q_{n-1}-q_n} - e^{q_n-q_{n+1}}$ : the equations of motion of the ring.*

*Proof.* Multiply out:

$$V_{n+1}L_n - L_nV_n = \begin{pmatrix} e^{q_n-q_{n+1}} - e^{q_{n-1}-q_n} & -p_n e^{-q_n} \\ -p_n e^{q_n} & 0 \end{pmatrix}, \quad \dot{L}_n = \begin{pmatrix} -\dot{p}_n & -\dot{q}_n e^{-q_n} \\ -\dot{q}_n e^{q_n} & 0 \end{pmatrix}.$$

Every  $\lambda$  has cancelled on the right. The off-diagonal entries agree iff  $\dot{q}_n = p_n$ , the diagonal ones iff  $\dot{p}_n$  has the stated value. ■

Why “curvature”? Draw the lattice of sites and times (Figure 10.4). Put  $L_n$  on each horizontal link, as transport from site  $n$  to  $n+1$ , and  $\mathbb{1} + V_n dt$  on each vertical link, as

transport from  $t$  to  $t + dt$ . Go round one plaquette, out along one pair of sides and back along the other:

$$(\mathbb{1} + V_n dt)^{-1} L_n(t + dt)^{-1} (\mathbb{1} + V_{n+1} dt) L_n(t) = \mathbb{1} + dt L_n^{-1} (V_{n+1} L_n - L_n V_n - \dot{L}_n) + O(dt^2). \quad (10.11)$$

This is the *holonomy* of the plaquette. In a flat geometry transport is path-independent and every small loop has trivial holonomy; so (10.10) says *the connection*  $(L_n, V_n)$  *on the space-time lattice is flat*. It is the discrete form of the zero-curvature condition

$$\partial_t U - \partial_x V + [U, V] = 0, \quad (10.12)$$

to which (10.10) reduces if  $L_n = \mathbb{1} + \Delta U(x)$ ,  $V_{n+1} = V + \Delta \partial_x V$  and terms of order  $\Delta^2$  are dropped. That equation, which integrates KdV and sine-Gordon, belongs to the sequel on field theory; it appears here once. The site-by-site transport, and its product round the ring, is the architecture [Lecture 17](#) quantises, with the Sklyanin bracket becoming the *RTT* relation.

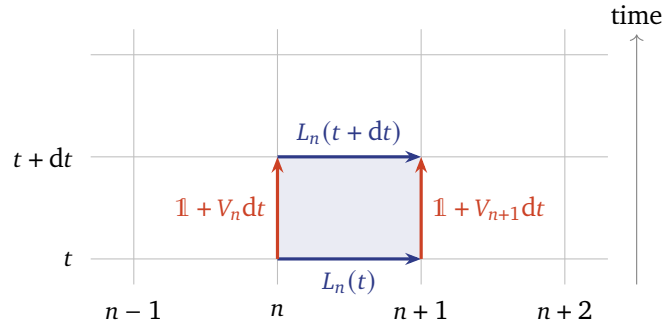


Figure 10.4: A plaquette of the space-time lattice. The two paths from lower left to upper right give  $(\mathbb{1} + V_{n+1} dt) L_n(t)$  and  $L_n(t + dt) (\mathbb{1} + V_n dt)$ ; zero curvature (10.10) says they agree to first order in  $dt$ .

## 10.4 The monodromy matrix around the ring

Transport all the way round:

$$T(\lambda) = L_N(\lambda) \cdots L_2(\lambda) L_1(\lambda), \quad t(\lambda) = \text{tr } T(\lambda), \quad \det T(\lambda) = 1. \quad (10.13)$$

These are the *monodromy matrix* and *transfer matrix* of [Definition 8.15](#), with the site matrices of (10.8). (The monodromy matrix is unrelated to the Hamiltonian monodromy of [Lecture 4](#); see [Remark 8.16](#).) By (10.8),  $\Phi_{N+1} = T(\lambda) \Phi_1$ :  $T$  is what a solution of (10.7) picks up in one circuit.

**Proposition 10.3 (The transfer matrix is conserved).**  $\dot{T}(\lambda) = [V_1(\lambda), T(\lambda)]$ . Hence  $t(\lambda)$  is conserved for every  $\lambda$ , and so is each of its coefficients.

*Proof.* Differentiate the product and use (10.10) on each factor. The sum telescopes:  $\dot{T} = V_{N+1} T - T V_1$ . On the ring  $q_{N+1} = q_1$  and  $q_N = q_0$ , so  $V_{N+1} = V_1$ . ■

This is a Lax equation for the  $2 \times 2$  matrix  $T(\lambda)$ , obtained from local data by transport. Periodicity is used exactly once, to close the telescope.

### What $t(\lambda)$ contains

Expand the product (10.13) as a sum over paths through the index values 1, 2 of the  $2 \times 2$  factors. The entry  $(L_n)_{22} = 0$  forbids a path to stay at 2, so each visit to 2 enters through  $(L_n)_{21} = -e^{q_n}$  and leaves at the next site through  $(L_{n+1})_{12} = e^{-q_{n+1}}$ : a factor  $-e^{q_n - q_{n+1}}$  in place of two factors  $\lambda - p$ . The trace closes the path, so the pair may straddle the join ( $n = N$ ,  $n + 1 = 1$ ). Hence

$$t(\lambda) = \sum_S \prod_{n \in S} (-e^{q_n - q_{n+1}}) \prod_{m \notin \bar{S}} (\lambda - p_m), \quad (10.14)$$

the sum over sets  $S$  of bonds, no two sharing a site, with  $\bar{S}$  the sites they cover. Writing  $t(\lambda) = \sum_{k=0}^N c_k \lambda^{N-k}$ ,

$$c_0 = 1, \quad c_1 = -\sum_n p_n = -F_1, \quad c_2 = \sum_{i < j} p_i p_j - \sum_n e^{q_n - q_{n+1}}, \quad H = \frac{1}{2} c_1^2 - c_2.$$

The  $c_k$  and the  $F_k = \text{tr } L^k$  of § 10.1 determine each other (Corollary 10.11); both are lists of the same  $N$  integrals.

### The Sklyanin bracket, verified

**Proposition 10.4** (The periodic Toda  $r$ -matrix). *Each site matrix satisfies the quadratic bracket (8.15),*

$$\{L_1(\lambda), L_2(\mu)\} = [r(\lambda - \mu), L_1(\lambda)L_2(\mu)], \quad r(\lambda) = -\frac{P}{\lambda}, \quad (10.15)$$

where  $L = L_n$ . Consequently  $\{t(\lambda), t(\mu)\} = 0$  for all  $\lambda, \mu$ , and the  $c_k$  are in involution.

*Proof.* Only  $L_{11} = \lambda - p$  involves  $p$ , so the only non-zero brackets pair it with an off-diagonal entry:  $\{\lambda - p, e^{-q}\} = -e^{-q}$  and  $\{\lambda - p, -e^q\} = -e^q$ , independent of  $\lambda$  and  $\mu$ . In § 8.2's notation,

$$\{L_1(\lambda), L_2(\mu)\} = e^{-q}(E_{12} \otimes E_{11} - E_{11} \otimes E_{12}) - e^q(E_{11} \otimes E_{21} - E_{21} \otimes E_{11}) = (E_{11} \otimes L^0 - L^0 \otimes E_{11})P,$$

with  $L^0 = L(0)$ ; the last step is (8.6) applied to each term. On the right of (10.15), write  $L(\lambda) = \lambda E_{11} + L^0$ . The permutation commutes with every  $A \otimes A$  (Proposition 8.4), which removes the  $\lambda\mu$  and the  $L^0 \otimes L^0$  terms, and it turns  $L^0 \otimes E_{11}$  into  $E_{11} \otimes L^0$ , so

$$[P, L(\lambda) \otimes L(\mu)] = (\lambda - \mu) [P, E_{11} \otimes L^0] = (\lambda - \mu)(L^0 \otimes E_{11} - E_{11} \otimes L^0)P.$$

Dividing by  $-(\lambda - \mu)$  gives the left-hand side. The chain is ultralocal, since  $L_n$  involves only  $(q_n, p_n)$ , so Theorem 8.17 lifts (10.15) to  $T$ , and Corollary 8.18 gives the involution. ■

This fixes the constant left free in Proposition 8.14: with this site matrix,  $c = -1$ . The other site matrix common in the literature,  $\begin{pmatrix} \lambda - p & e^q \\ -e^{-q} & 0 \end{pmatrix}$ , gives the reversed potential  $e^{q_{n+1} - q_n}$  and  $c = +1$  (scripts/check-rmatrix.py); each sign belongs to its pairing. The open chain (§ 9.4) needed a constant  $r$  in the linear bracket; the ring's is quadratic, the form that composes.

### Integrability, and the tori of the ring

**Theorem 10.5** (The periodic Toda chain is Liouville integrable). *On  $\mathbb{R}^{2N}$  the coefficients  $c_1, \dots, c_N$  of  $t(\lambda)$  are in involution and independent, and  $H = \frac{1}{2} c_1^2 - c_2$ . The periodic Toda*

chain satisfies [Definition 4.6](#).

*Proof.* Involution is [Proposition 10.4](#). For independence, read (10.14):  $c_k = (-1)^k e_k(p) + R_k$ , where  $e_k$  is the  $k$ -th elementary symmetric polynomial of the momenta, and every term of  $R_k$  has a factor  $e^{q_n - q_{n+1}}$  and degree at most  $k - 2$  in  $p$  (each bond removes two factors  $\lambda - p$ ). In the Jacobian  $\det(\partial c_k / \partial p_n)$  the part of top degree in  $p$  therefore comes from the  $e_k$  alone, and it is  $\pm \prod_{i < j} (p_j - p_i)$ , the Vandermonde determinant (a classical identity for the elementary symmetric polynomials; the same determinant opened [§ 9.3](#), and `scripts/check-110-algebra.py` checks the whole statement for  $N = 3, 4$ ). So at fixed  $q$  the Jacobian is a non-zero polynomial in  $p$ , which cannot vanish on any open set. Where it does not vanish, the  $dc_k$  are independent. That set is open, and it meets every open set, so it is dense. ■

This is weaker than [Theorem 9.3](#), and has to be: independence does fail somewhere, for instance at the equilibrium, where every flow is the drift. [Definition 4.6](#) asks only for a dense open set. The hypothesis audit of [§ 9.5](#) then runs as follows.

**Proposition 10.6** (Tori of the ring · proved modulo [Lemma 4.11](#) and [Theorem 1.10](#)). *Let  $\Sigma$  be a connected component of a level set of  $(c_1, \dots, c_N)$  on which  $dc_1, \dots, dc_N$  are independent at every point, and let  $Q = \frac{1}{N} \sum_n q_n$  be the centre of mass. Then  $\Sigma \cong T^{N-1} \times \mathbb{R}$ , and each slice  $\Sigma \cap \{Q = \text{const}\}$  is a torus  $T^{N-1}$ .*

The proof has four steps. *Bounded:* on a level set the spectrum of  $L$  is fixed, so every  $|p_n|$  and  $a_n$  is bounded above, and then (10.2) bounds every  $a_n$  below; at fixed  $Q$  the slice is compact. *Complete:* as in [Proposition 9.6](#). *Cylinder:*  $L^4$ 's stages give  $\Sigma \cong \mathbb{R}^N / \Lambda$ ,  $\Lambda$  discrete, as in [Corollary 9.7](#). *Drift:*  $Q$  moves at constant rate along every flow, so periods lie in an  $(N - 1)$ -dimensional subspace, on which compactness of the slice forces a full lattice.

Reading

*Proof of Proposition 10.6. Bounded.* On  $\Sigma$  the polynomial  $t$  is fixed, so is  $\det(\lambda - L) = t(\lambda) - 2$  ([Corollary 10.11](#), which does not use this proposition), so the eigenvalues of  $L$  are fixed and by (L2) of [§ 9.2](#) every  $|p_n|$  and  $a_n$  is at most a constant  $C$ . By (10.2) every  $a_n = 1 / \prod_{m \neq n} a_m \geq C^{1-N}$  as well. So each  $q_n - q_{n+1}$  is bounded; at fixed  $Q$  so is each  $q_n$ . A component of a regular level set is closed, so  $\Sigma \cap \{Q = \text{const}\}$  is closed and bounded: compact.

*Complete.* The components of each  $X_{c_k}$  are polynomials in  $p_n$  and  $e^{q_n - q_{n+1}} = a_n^2$ , bounded on  $\Sigma$ , so the proof of [Proposition 9.6](#) applies with the  $c_k$  in place of the  $F_k$ .

*Cylinder.* As in [Corollary 9.7](#), stages (i)–(iii) of [§ 4.4](#) and the first half of stage (iv) make  $\Sigma$  the orbit of an action of  $\mathbb{R}^N$  with a discrete stabiliser  $\Lambda$ , so  $\Sigma \cong \mathbb{R}^N / \Lambda$ .

*The drift.* Shifting every  $p_n$  by  $s$  is the same as shifting  $\lambda$  by  $-s$  in (10.14), so  $\sum_n \partial c_k / \partial p_n$  is  $-(N - k + 1)c_{k-1}$ , constant on  $\Sigma$ . Hence  $Q$  changes at a constant rate along each flow, and  $Q(\text{time-}s \text{ point}) = Q(x_0) + \ell(s)$  for a linear map  $\ell : \mathbb{R}^N \rightarrow \mathbb{R}$ , not zero since  $\ell(e_1) = -1$ . A period does not move  $Q$ , so  $\Lambda \subset \ker \ell \cong \mathbb{R}^{N-1}$ , and  $\Sigma \cap \{Q = Q(x_0)\} \cong \ker \ell / \Lambda$ . That is compact, so by [Lemma 4.11](#)  $\Lambda$  is a lattice of full rank in  $\ker \ell$ : the slice is  $T^{N-1}$  and  $\Sigma \cong T^{N-1} \times \mathbb{R}$ . ■

So the ring delivers what [§ 9.8](#) promised. The open chain's particles could separate; on the ring (10.2) forbids it. For  $N = 3$  these 2-tori are the closed curves of [Figure 6.9](#).

## 10.5 The spectral curve, bands and gaps

Now put the two constructions side by side: the  $N \times N$  matrix  $L(z)$  of § 10.2 and the  $2 \times 2$  matrix  $T(\lambda)$  of § 10.4. They came from the same recursion.

**Proposition 10.7** (Bloch waves are eigenvectors of the monodromy matrix). *For  $z \neq 0$  and any  $\lambda$ , the map  $\psi \mapsto \Phi_1$  is a linear isomorphism from the eigenspace of  $L(z)$  for the eigenvalue  $\lambda$  onto the eigenspace of  $T(\lambda)$  for the eigenvalue  $z$ . In particular  $\lambda$  is an eigenvalue of  $L(z)$  if and only if  $z$  is an eigenvalue of  $T(\lambda)$ , and then*

$$z + \frac{1}{z} = t(\lambda). \quad (10.16)$$

*Proof.* Given  $L(z)\psi = \lambda\psi$ , extend  $\psi$  to all  $n$  by (10.5); it solves (10.4) for every  $n$ . Gauge to  $\phi_n = e^{-q_n/2}\psi_n$  and form  $\Phi_n$ . By (10.8),  $\Phi_{N+1} = T(\lambda)\Phi_1$ , and by periodicity of the  $q_n$ ,  $\Phi_{N+1} = z\Phi_1$ . If  $\Phi_1 = 0$  then  $\phi_0 = \phi_1 = 0$ , and (10.7) run both ways gives  $\phi \equiv 0$ ; so the map is injective. Conversely, an eigenvector  $\Phi_1$  of  $T(\lambda)$  generates  $\Phi_n = L_{n-1} \cdots L_1 \Phi_1$  (and, using  $L_n^{-1}$ , which exists since  $\det L_n = 1$ , for  $n < 1$ ), whose first components solve (10.7), with  $\Phi_{n+N} = z\Phi_n$  because the site matrices are periodic; ungauging gives a Bloch solution of (10.4), which is an eigenvector of  $L(z)$  by § 10.2. Finally  $\det T = 1$ , so the eigenvalues of  $T$  are  $z$  and  $1/z$ , and their sum is  $t(\lambda)$ . ■

**Definition 10.8** (Spectral curve). The *spectral curve* of the periodic Toda chain is  $\Gamma = \{(\lambda, z) \in \mathbb{C} \times \mathbb{C}^\times : z + 1/z = t(\lambda)\}$ .

By Proposition 10.3 the curve does not move. Its coefficients are the integrals, and it holds all of them at once. Spectral curves of this kind, and their quantum counterparts, return in Lecture 23 as a frontier.

### Bands and gaps

Take  $\lambda$  real. Then  $t(\lambda)$  is real and the two roots of  $z^2 - t(\lambda)z + 1 = 0$  are

$$z_{\pm} = \frac{t(\lambda) \pm \sqrt{t(\lambda)^2 - 4}}{2}, \quad z_+ z_- = 1.$$

- If  $|t(\lambda)| \leq 2$ , they are  $e^{\pm i\theta}$  with  $2 \cos \theta = t(\lambda)$ : the recursion has bounded Bloch solutions, the electron propagates, and  $\lambda$  is in a *band*.
- If  $|t(\lambda)| > 2$ , they are real, one of modulus above 1 and one below: every non-zero solution is unbounded along the crystal, and  $\lambda$  is in a *gap*.
- The *band edges* are where  $t(\lambda) = \pm 2$ , that is  $z = \pm 1$ : by Proposition 10.7, the eigenvalues of the periodic matrix  $L(1) = L$  and of the antiperiodic one  $L(-1)$ . Both are real symmetric, so every band edge is real; § 10.6 shows that there are  $2N$  of them, counted with multiplicity.

Figure 10.5 shows  $t(\lambda)$  for a six-particle ring:  $N = 6$  bands, where it crosses the strip  $|t| \leq 2$  monotonically, and  $N - 1 = 5$  gaps, where it turns round outside it (Corollary 10.11 proves this is general). The momenta keep changing; the band edges do not.

At the equilibrium, all  $q_n$  equal (equally spaced particles, by the footnote) and all  $p_n = b$ , every  $a_n = 1$  and one finds  $t(\lambda) = 2T_N((\lambda - b)/2)$ , with  $T_N$  the Chebyshev polynomial: it touches  $\pm 2$  at every turning point, all gaps are closed, and the spectrum is the single band  $[b - 2, b + 2]$  of the free tight-binding model. Outside the loci where a gap collapses, which

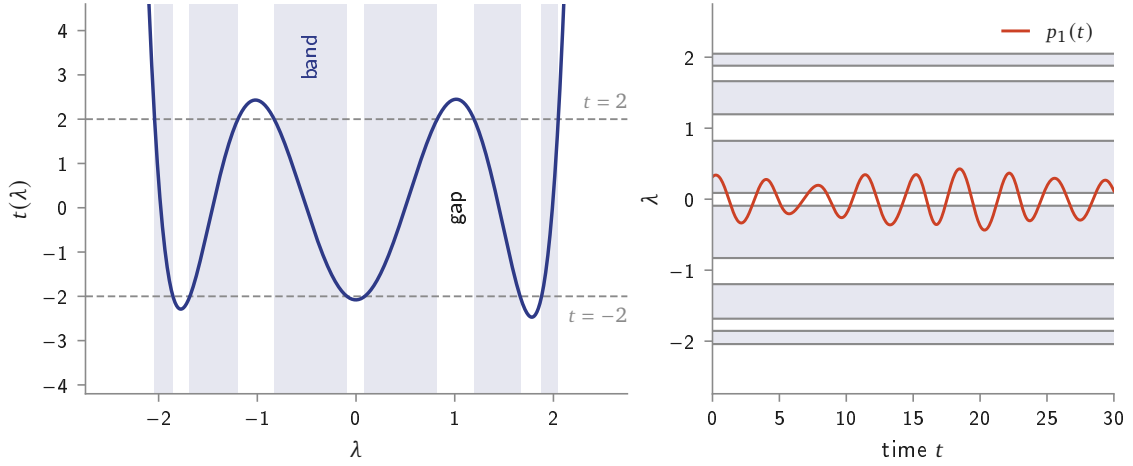


Figure 10.5: Left:  $t(\lambda)$  for a state of the  $N = 6$  ring, bands shaded. Right: the  $2N$  band edges (the spectra of  $L(\pm 1)$ ) along the motion, constant while  $p_1(t)$  oscillates (scripts/figures/toda\_bands.py asserts the edges constant to  $10^{-8}$  and equal to the roots of  $t \mp 2$ ).

have codimension two (Henrici and Kappeler, §2 · *cited*), all  $N - 1$  gaps are open; a special excitation can leave some closed (a state repeating with a shorter period does).

#### Tutorial

**Example 10.9 (All gaps closed: the equilibrium).** At  $q_1 = \dots = q_N$  and  $p_1 = \dots = p_N = b$  every  $a_n = 1$ , and  $L(z)$  is the free tight-binding model. Its Bloch waves are plane waves  $\psi_n = e^{ikn}$ , with  $e^{ikN} = z$  and energy  $\lambda = b + 2 \cos k$ . Putting  $\lambda - b = 2 \cos k$ ,

$$t(\lambda) = z + \frac{1}{z} = 2 \cos Nk = 2 T_N\left(\frac{\lambda - b}{2}\right),$$

with  $T_N$  the Chebyshev polynomial,  $T_N(\cos k) = \cos Nk$ . It touches  $\pm 2$  at every turning point, so all  $N - 1$  gaps are closed and the spectrum is one band  $[b - 2, b + 2]$ . This state is the ring at rest in its centre-of-mass frame, drifting with speed  $b$ . Outside codimension-two loci every gap is open (Henrici–Kappeler, *cited*), but a special excitation need not open them all (for  $N = 4$ , a state repeating with period 2 has  $t = d^2 - 2$ ,  $d$  the two-site trace, and two of its three gaps stay closed). A gap is the spectral trace of motion relative to the drift, as § 10.7 makes quantitative. At this state the  $dc_k$  are dependent. By the cyclic symmetry every  $X_{c_k}$  moves all particles alike, and it changes no momentum, since  $\sum_n \partial c_k / \partial q_n = 0$  (the  $c_k$  see only differences of positions) and the  $N$  terms are equal. So every flow is the drift, the  $X_{c_k}$  span one direction, and for  $N \geq 2$  the  $dc_k$  cannot be independent.

### The curve as a two-sheeted cover

Above each  $\lambda$  lie two points of  $\Gamma$ ,  $z_+$  and  $z_- = 1/z_+$ , which coincide exactly at the band edges. So  $\Gamma$  is two copies of the  $\lambda$ -plane (the *sheets*) glued along cuts joining the band edges in pairs; crossing a cut exchanges  $z$  and  $1/z$  (Figure 10.6). With all gaps open,  $2N$  branch points and  $N$  cuts give a closed surface with  $N - 1$  handles: *genus*  $N - 1$  (Babelon–Bernard–Talon, §6.2 · *cited*), the dimension of the tori of Proposition 10.6. § 10.7 says why.

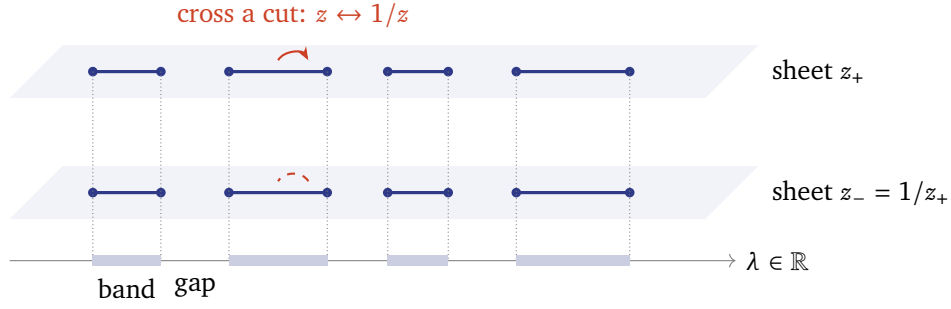


Figure 10.6: The spectral curve as a two-sheeted cover of the  $\lambda$ -plane, drawn over the real axis for  $N = 4$ . The cuts (thick) run along the bands and end at the  $2N$  branch points, where  $z_+ = z_- = \pm 1$ ; crossing a cut changes sheet. Two sheets glued along  $N$  cuts make a surface with  $N - 1$  handles.

*A warning on names.* Babelon–Bernard–Talon exchange the letters: their  $\lambda$  is our  $z$ , their  $\mu$  our  $\lambda$ , their  $2t(\mu)$  our  $t(\lambda)$ , and their  $n + 1$  particles our  $N$  (so their genus  $n$  is our  $N - 1$ ).

## 10.6 Spectral duality: one curve, read two ways

**Proposition 10.7** matched the two eigenvalue problems as sets. The polynomial identity is sharper.

**Theorem 10.10 (Spectral duality).** For  $N \geq 3$  and all  $\lambda \in \mathbb{C}$ ,  $z \neq 0$ ,

$$\det(\lambda - L(z)) = t(\lambda) - \left(z + \frac{1}{z}\right) \quad (10.17)$$

The left side is the characteristic polynomial of the  $N \times N$  Lax matrix, whose spectral parameter is  $z$ ; the right side is the characteristic equation of the  $2 \times 2$  monodromy matrix, whose spectral parameter is  $\lambda$ . They define the same curve  $\Gamma$ .

*Proof.* Expand the determinant over permutations  $\sigma$ . The matrix  $\lambda - L(z)$  is tridiagonal apart from its corners,  $-a_N z$  at  $(N, 1)$  and  $-a_N/z$  at  $(1, N)$ . A term using the  $(N, 1)$  corner and not the  $(1, N)$  one must map  $N \mapsto 1$  and, to stay within the band,  $n \mapsto n + 1$  for every  $n$ : the  $N$ -cycle, of sign  $(-1)^{N-1}$ , whose term is  $(-1)^{N-1} \prod_n (-a_n) z = -z$  by (10.2). Likewise the other corner alone gives  $-1/z$ , and every other term uses both corners or neither and does not involve  $z$ . So  $\det(\lambda - L(z)) = A(\lambda) - (z + 1/z)$  with  $A$  independent of  $z$ . Fix  $\lambda$  and choose  $z$  with  $z + 1/z = A(\lambda)$ . Then  $\det(\lambda - L(z)) = 0$ , so  $\lambda$  is an eigenvalue of  $L(z)$ , so by **Proposition 10.7**  $z + 1/z = t(\lambda)$ . Hence  $A(\lambda) = t(\lambda)$ . ■

The merged  $N = 2$  matrix satisfies (10.17) too (scripts/check-l10-algebra.py,  $N = 2, \dots, 5$ , symbolic).

**Two spectral parameters, one curve.**

- $z$ , the *Bloch–Floquet multiplier*, lives in the corners of the  $N \times N$  matrix  $L(z)$ . Fix  $z$ :  $N$  points of  $\Gamma$ , the eigenvalues of  $L(z)$ .  $\Gamma$  is an  $N$ -sheeted cover of the  $z$ -plane.
- $\lambda$ , the *energy* of the Bloch wave, is the spectral parameter of the  $2 \times 2$  site matrices. Fix  $\lambda$ : 2 points of  $\Gamma$ , the eigenvalues  $z, 1/z$  of  $T(\lambda)$ .  $\Gamma$  is a *two-sheeted* cover of the  $\lambda$ -plane.
- The eigenvalue of one Lax matrix is the spectral parameter of the other. The Flaschka normalisation was chosen so that this holds with no factor.

Figure 10.7 draws the  $N$ -sheeted reading over the unit circle  $|z| = 1$ : the familiar band diagram of a crystal. A vertical line meets it in  $N$  points, a horizontal line inside a band in 2.

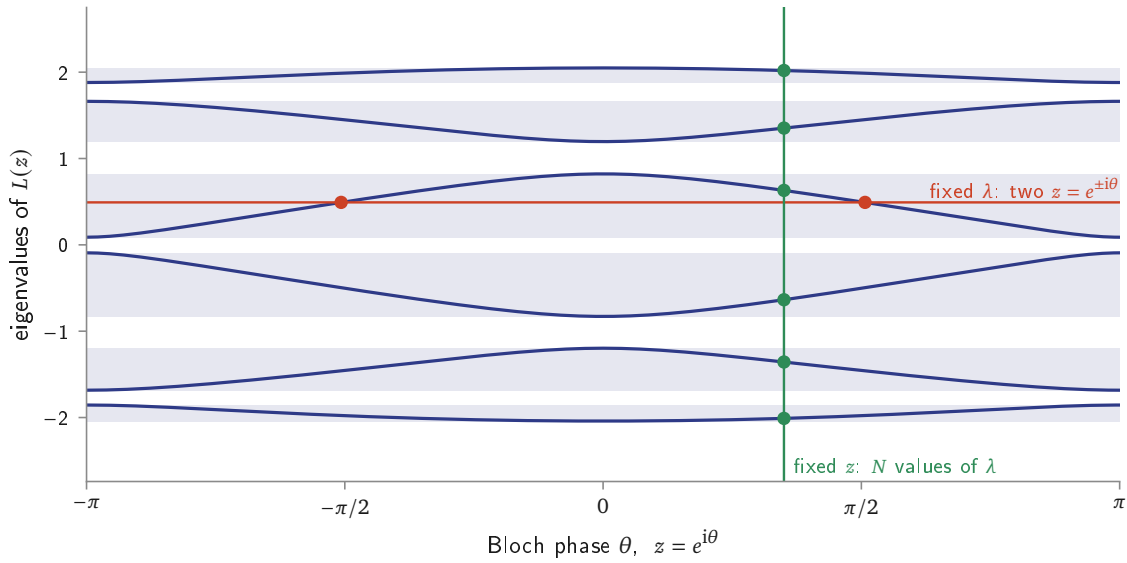


Figure 10.7: The state of Figure 10.5: eigenvalues of  $L(e^{i\theta})$  against the Bloch phase. Each of the  $N = 6$  branches sweeps one band. Fixed  $z$  (green):  $N$  values of  $\lambda$ ; fixed  $\lambda$  (red): two values of  $z$  (toda\_bands.py asserts both, and (10.17) at sampled points).

Duality pays off at once.

**Corollary 10.11 (Bands and gaps).** (i)  $\det(\lambda - L) = t(\lambda) - 2$ , so the coefficients  $c_k$  and the integrals  $F_k = \text{tr } L^k$  of § 10.1 determine each other, and the  $F_k$  are in involution. (ii)  $t'(\lambda) \neq 0$  wherever  $|t(\lambda)| < 2$ . Hence the real line consists of exactly  $N$  bands, on each of which  $t$  runs monotonically from  $\pm 2$  to  $\mp 2$ , separated by  $N - 1$  gaps, some of which may be closed.

*Proof.* (i) is (10.17) at  $z = 1$ , and Newton's identities express the power sums  $\text{tr } L^k$  of the eigenvalues through the coefficients of the characteristic polynomial, and back. (ii) At a point of a band interior,  $t(\lambda_0) = 2 \cos \theta$  with  $z = e^{i\theta} \neq 1/z$ , so the eigenvalue  $\lambda_0$  of the Hermitian  $L(z)$  is simple by Proposition 10.7, and it is a simple root of  $t(\lambda) - 2 \cos \theta$  by (10.17). The count of bands follows (reading). ■

## Reading

*Proof of (ii) in full.* Let  $t(\lambda_0) = 2 \cos \theta$  with  $0 < \theta < \pi$ , and  $z = e^{i\theta}$ . Since  $z \neq 1/z$ , the eigenspace of  $T(\lambda_0)$  for  $z$  is one-dimensional, so by [Proposition 10.7](#) so is that of  $L(z)$  for  $\lambda_0$ .  $L(z)$  is Hermitian, so  $\lambda_0$  is a simple root of  $\det(\lambda - L(z)) = t(\lambda) - 2 \cos \theta$ , and  $t'(\lambda_0) \neq 0$ . So on each maximal open interval where  $|t| < 2$  the polynomial is strictly monotone; the interval is bounded, since  $t \rightarrow \pm\infty$ , and  $t = +2$  at one end and  $-2$  at the other. Each such interval therefore contains exactly one solution of  $t(\lambda) = 2 \cos \theta$  for each  $\theta \in (0, \pi)$ . There are exactly  $N$  solutions, the eigenvalues of  $L(e^{i\theta})$ , all simple, so there are exactly  $N$  such intervals: the bands. Between consecutive bands lie the  $N - 1$  gaps, closed where two bands touch. ■

## 10.7 What $z$ does and does not buy

*An honest remark:*  $z$  gives no new integrals. Expanding (10.17) with Newton's identities,

$$\operatorname{tr} L(z)^k = \operatorname{tr} L^k \quad (k < N), \quad \operatorname{tr} L(z)^N = \operatorname{tr} L^N + N \left( z + \frac{1}{z} - 2 \right).$$

The  $z$ -dependence is carried by  $a_1 \cdots a_N$ , a constant by (10.2):  $L(z)$  holds exactly the integrals  $L$  had. Contrast Manakov's parameter in § 7.9, which recovered a missed integral.

*Where the spectral parameter is indispensable.* In the  $2 \times 2$  formulation: all  $N$  integrals are the coefficients of one  $2 \times 2$  trace, and without  $\lambda$  the site matrices carry nothing. In the bracket: (10.15) compares  $\lambda$  with  $\mu$ , and without it there is no composition along the ring; [Lecture 17](#) quantises exactly this, with  $T(\lambda)$  operator-valued,  $RTT$  in place of (10.15), and  $t(\lambda)$  generating a spin-chain Hamiltonian. And in the geometry: the actions of [Proposition 10.6](#) are periods of the curve.

### Action variables from the curve

When all  $N - 1$  gaps are open, the curve has  $N - 1$  closed loops that stay over the real axis (a closed gap collapses its loop to a point): above the  $j$ -th gap, both roots  $z_{\pm}$  are real, and the loop runs along the gap on one sheet and back on the other, crossing between them at the two band edges ([Figure 10.8](#)). Call it  $\alpha_j$ .

**Proposition 10.12** (Actions from the gaps · cited). *On a regular torus of [Proposition 10.6](#) with all  $N - 1$  gaps open, the loops  $\alpha_j$  correspond to a basis  $\gamma_j$  of cycles of the torus  $T^{N-1}$  in the sense of [Lecture 4](#), and the action variables (5.1) of the relative motion are*

$$I_j = \frac{1}{2\pi} \oint_{\alpha_j} \lambda \, d \ln z = \frac{1}{\pi} \int_{\text{gap } j} \operatorname{arccosh} \frac{|t(\lambda)|}{2} \, d\lambda, \quad j = 1, \dots, N - 1. \quad (10.18)$$

(Flaschka and McLaughlin 1976; in this normalisation, Henrici and Kappeler 2008, Definition 3.1 and Appendix A. The separated variables behind it are Babelon–Bernard–Talon, §§6.5–6.6.)

The second form follows by integrating by parts, with  $\ln |z_{\pm}| = \pm \operatorname{arccosh}(|t|/2)$  over a gap and  $\alpha_j$  oriented so that  $I_j \geq 0$ . A closed gap has zero action; a wide gap means a large amplitude. The proof needs separated variables and the Riemann surface of  $\Gamma$ , which this course does not build. `scripts/check-l10-algebra.py` checks (10.18) for  $N = 4, 5$  near

equilibrium, where  $H - N - F_1^2/2N = \sum_j \nu_j I_j$  to leading order,  $\nu_j = 2 \sin(\pi j/N)$  (relative error below  $10^{-5}$ ), and the tutorial does  $N = 2$  by hand.

Tutorial

**Example 10.13 (The two-particle ring).** Take  $N = 2$ , with  $x = q_1 - q_2$  and  $p_x = \frac{1}{2}(p_1 - p_2)$ , a canonical pair, and the mean momentum  $\bar{p} = \frac{1}{2}(p_1 + p_2)$  (not  $P$ , the permutation operator). Then  $H = \bar{p}^2 + E$  with relative energy  $E = p_x^2 + 2 \cosh x \geq 2$ : a particle in a cosh well. From (10.14), with  $p_1 p_2 = \bar{p}^2 - p_x^2$ ,

$$t(\lambda) = (\lambda - p_1)(\lambda - p_2) - e^x - e^{-x} = u^2 - E, \quad u = \lambda - \bar{p}.$$

There is one finite gap,  $|u| < \sqrt{E - 2}$ , where  $t < -2$ , and there  $\operatorname{arccosh}(|t|/2)$  is the  $x_0 \geq 0$  with  $2 \cosh x_0 = E - u^2$ . So the right side of (10.18) is  $\frac{1}{\pi} \int x_0(u) du$ : the area of the region  $\{u^2 + 2 \cosh x \leq E\}$  in the  $(u, x)$ -plane, divided by  $2\pi$ . Rename  $u$  as  $p_x$ : it is the area enclosed by the orbit in the  $(x, p_x)$ -plane over  $2\pi$ , which is  $\frac{1}{2\pi} \oint p_x dx$ . On the slice  $Q = \text{const}$  we have  $p \cdot dq = 2\bar{p} dQ + p_x dx = p_x dx$ , so this is the action of Definition 5.1. The same region, sliced two ways.

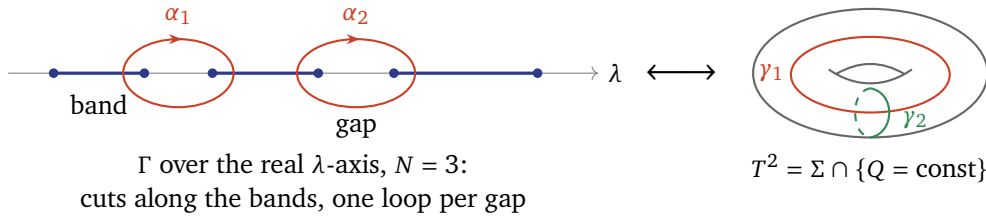


Figure 10.8: Left: for  $N = 3$ , two gaps and two real loops  $\alpha_1, \alpha_2$  on the curve, each along a gap on one sheet and back on the other. Right: the relative torus  $T^2$  of Proposition 10.6 with a basis of cycles. Proposition 10.12 (cited) matches the two.

This is the sense in which the genus of  $\Gamma$  equals the dimension of the tori: each open gap is a handle of the curve and a circle of the torus, and its width measures how hard that circle is excited. Theta functions on  $\Gamma$  then give the motion itself in closed form (Babelon–Bernard–Talon, §6.4 · cited), which we do not pursue.

### Summary.

- *The ring* (10.1): Flaschka's  $L$  with  $a_N$  in the corners,  $\dot{L} = [M, L]$ , and  $a_1 \cdots a_N = 1$  (10.2). The whole ring still drifts.  $N = 2$  is degenerate for  $L$ .
- *Bloch–Floquet*: imposing  $\psi_{n+N} = z\psi_n$  on the unrolled recursion gives  $L(z)$  (10.6), with a Lax pair for every  $z$  (Proposition 10.1).
- *Zero curvature*: the gauge  $\phi_n = e^{-q_n/2}\psi_n$  turns the recursion into  $\Phi_{n+1} = L_n(\lambda)\Phi_n$  with the local site matrix (10.8), and  $\dot{L}_n = V_{n+1}L_n - L_nV_n$  is equivalent to the equations of motion (Proposition 10.2): every plaquette has trivial holonomy.
- *Monodromy*:  $T = L_N \cdots L_1$  obeys  $\dot{T} = [V_1, T]$ ; the quadratic bracket holds with  $r(\lambda) = -P/\lambda$  (Proposition 10.4), so the coefficients of  $t(\lambda)$  commute. The ring is Liouville integrable (Theorem 10.5), and its regular level sets are  $T^{N-1} \times \mathbb{R}$  (Proposition 10.6).
- *The curve*  $z + 1/z = t(\lambda)$  (Proposition 10.7): bands where  $|t| \leq 2$ , gaps where  $|t| > 2$ ,  $N$  bands and  $N - 1$  gaps (Corollary 10.11); genus  $N - 1$ .

- *Spectral duality* ([Theorem 10.10](#)):  $\det(\lambda - L(z)) = t(\lambda) - z - 1/z$ , one curve read as an  $N$ -sheeted cover in  $z$  and a two-sheeted cover in  $\lambda$ .
- *Actions* are integrals over the gaps ([Proposition 10.12](#), cited;  $N = 2$  by hand in the tutorial).

**Tutorial.** Exercise Sheet 10.

## References

Babelon, Bernard and Talon, *Introduction to Classical Integrable Systems*, ch. 6: §6.1 (the closed chain and its Lax pair), §6.2 (the spectral curve and its genus), §6.3 (Bloch waves), §§6.5–6.6 (separated variables, the  $2 \times 2$  monodromy matrix); Toda, *Theory of Nonlinear Lattices*, ch. 4 (the periodic lattice); H. Flaschka and D. W. McLaughlin, *Prog. Theor. Phys.* 55 (1976) 438 (action variables of the periodic lattice from the gaps); A. Henrici and T. Kappeler, *Global action-angle variables for the periodic Toda lattice*, arXiv:0802.4032 (§2, the collapsed-gap loci; the actions in this normalisation); P. van Moerbeke, *Invent. Math.* 37 (1976) 45 (the spectrum of periodic Jacobi matrices). **Conventions.** Babelon–Bernard–Talon exchange the letters: their  $\lambda$  is our  $z$ , their  $\mu$  our  $\lambda$ , their  $2t$  our  $t$ ; their chain has  $n + 1$  particles, and their bracket  $\{p, q\} = \frac{1}{2}$  rescales  $q$  relative to ours.